UNIVERSITY OF BIRMINGHAM

LM Big Data Management in Finance (36320)

Birmingham Business School

Week-3

Fundamentals of Machine Learning

Dr. Animesh Acharjee

Module Lead

Agenda

- What is machine learning
- Types of machine learning
- Fundamentals of machine learning
- Validation matrices
- Why penalization methods required?
- What is penalization?

What is machine learning

- Machine learning, a branch of artificial intelligence, concerns the construction and study of systems that can learn from data (http://en.wikipedia.org/wiki/Machine_learning)

- Why we should use machine learning?
  - Machine learning methods are not based on statistical assumptions such as normality, independence etc.

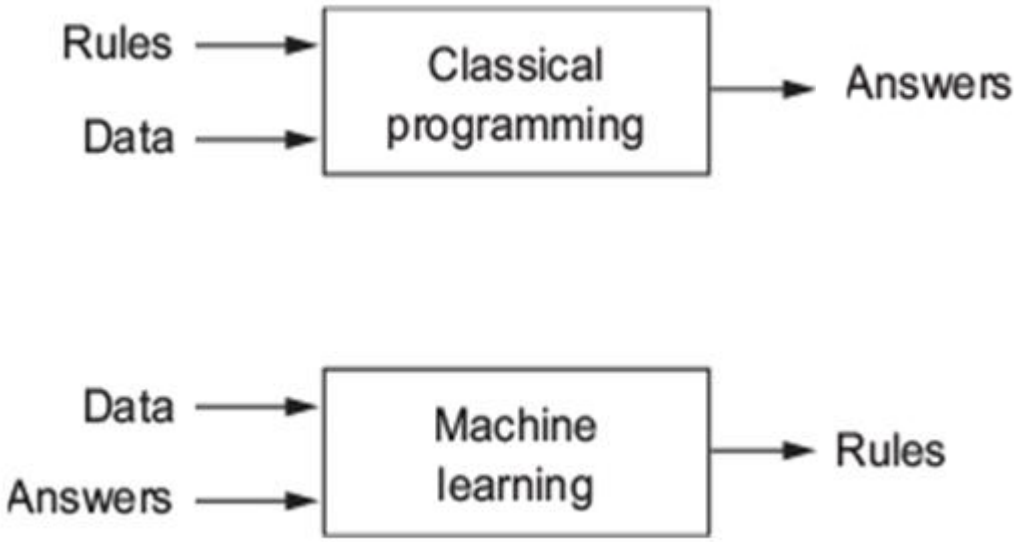
- All machine learning methods are non-parametric but reverse is *not* always true

- Some machine learning algorithms are inspired by biological phenomena. Ex: Genetic algorithm, ant colony optimization

Classical vs. machine learning models

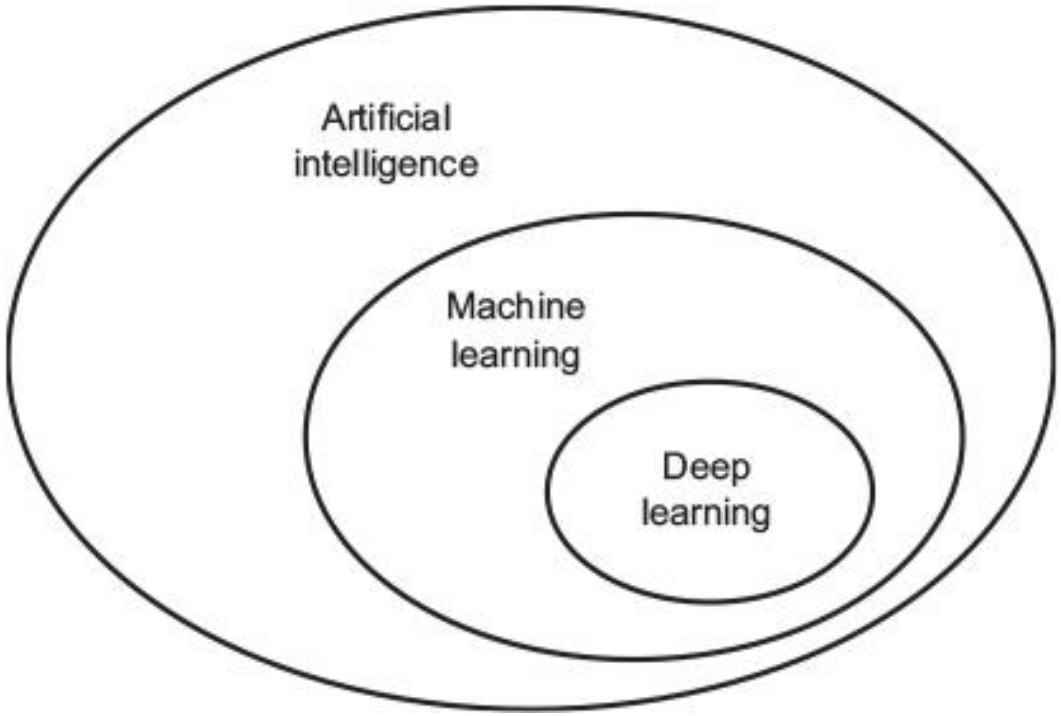
- Classical models are presented with rules and data to get answers

- Machine learning systems is trained rather explicitly programmed

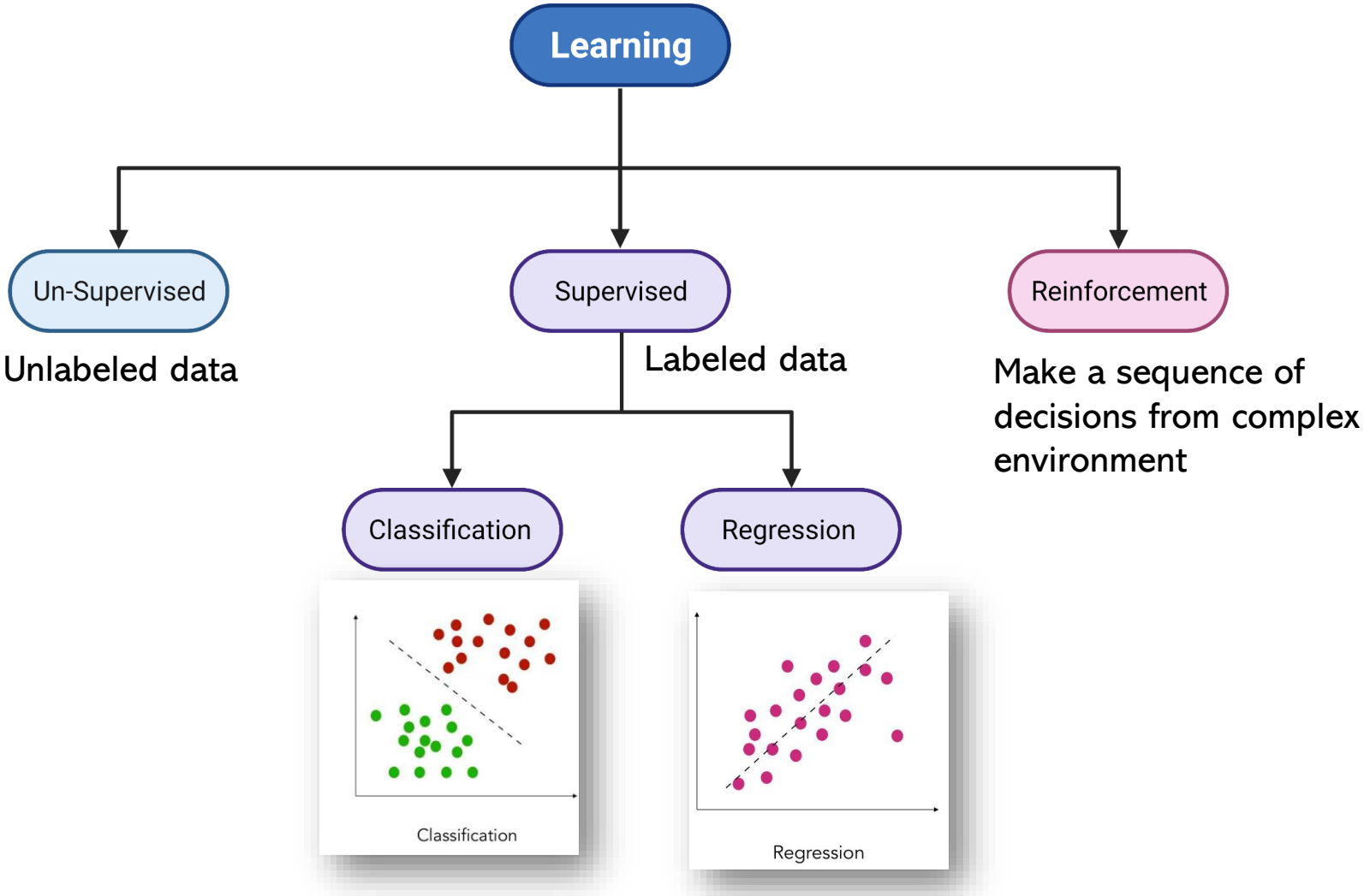


Artificial intelligence, machine learning, and deep learning

- Artificial Intelligence (AI)
  - Involves machines that can perform tasks/job that are characteristic of human intelligence.
  - Might involve some machine learning
- Machine Learning (ML)
  - Development of algorithms, which improve by presenting data
- Deep Learning
  - Part of machine learning and AI
  - Deep models
  - Based on artificial neural networks.



Types of Learning



Algorithm

- Machine learning tasks are typically classified into three broad categories

Supervised learning

- The computer/algorithm is presented with example inputs and their desired outputs, given by a “teacher” (user), and the goal is to learn a general rule that maps inputs to outputs

Unsupervised learning

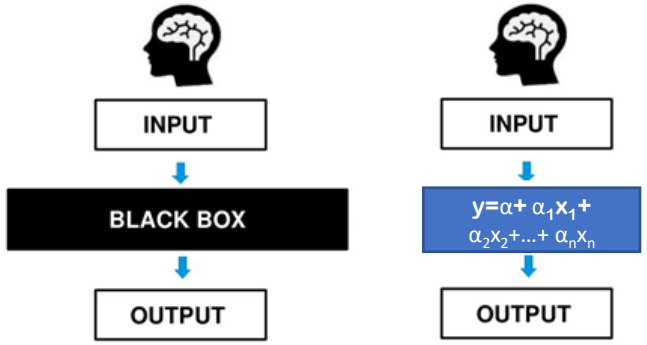
- No labels are given to the learning algorithm, leaving it on its own to find structure in its input
- Unsupervised learning can be a goal in itself (discovering hidden patterns in data)

Reinforcement learning

- A computer program interacts with a dynamic environment in which it must perform a certain goal (such as driving a vehicle), without a teacher explicitly telling it whether it has come close to its goal or not

Machine learning vs. Statistics

Machine learning	Statistics
network, graphs	model
weights	parameters
learning	fitting
generalization	test set performance
supervised learning	regression/classification
unsupervised learning	density estimation, clustering



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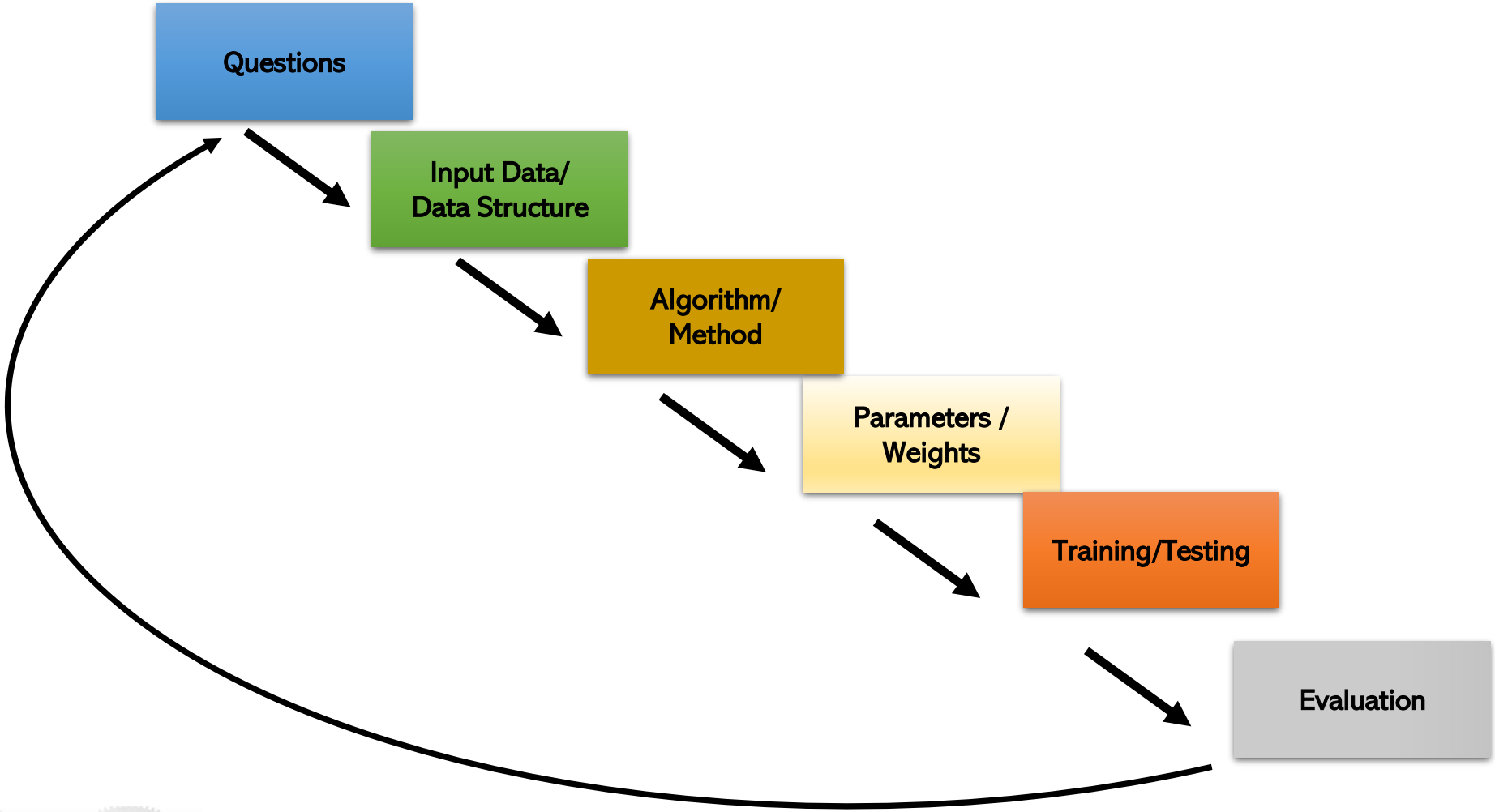
Work Flow of Machine Learning Methods



Questions

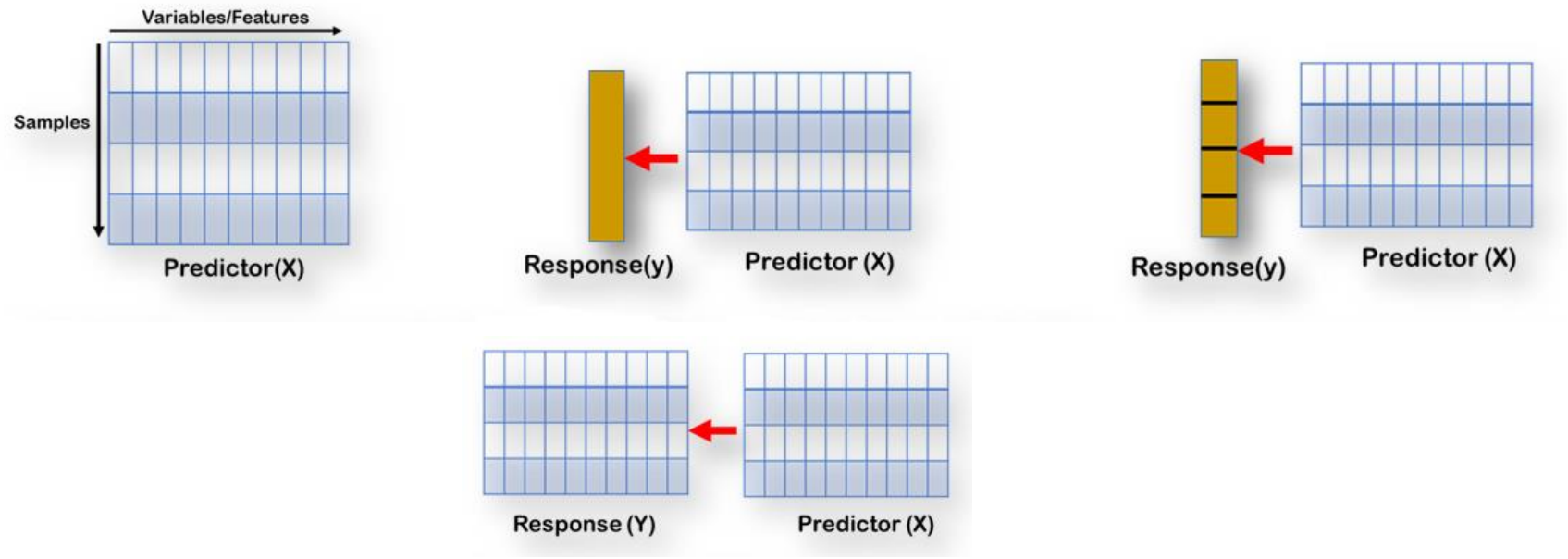
Questions are usually in two categories

- Observation/individual related
  - How observations are clustered? For example: Credit card fraud or not
  - Outliers
  - Missing values

- Feature/Variable related
  - How variables/features are clusters?
  - Which variables are making differences among many classes?
  - Which variables are correlated ?
  - Is it possible to identify them in the data set?
  - Are those variables being predictive? Can be used as a stratification in different location?
  - Are those variables being stable?
  - Are those variables statistically significant?

Input Data

- Input data can have different structures
- It is very important to understand what type of question is being asked and does data support that



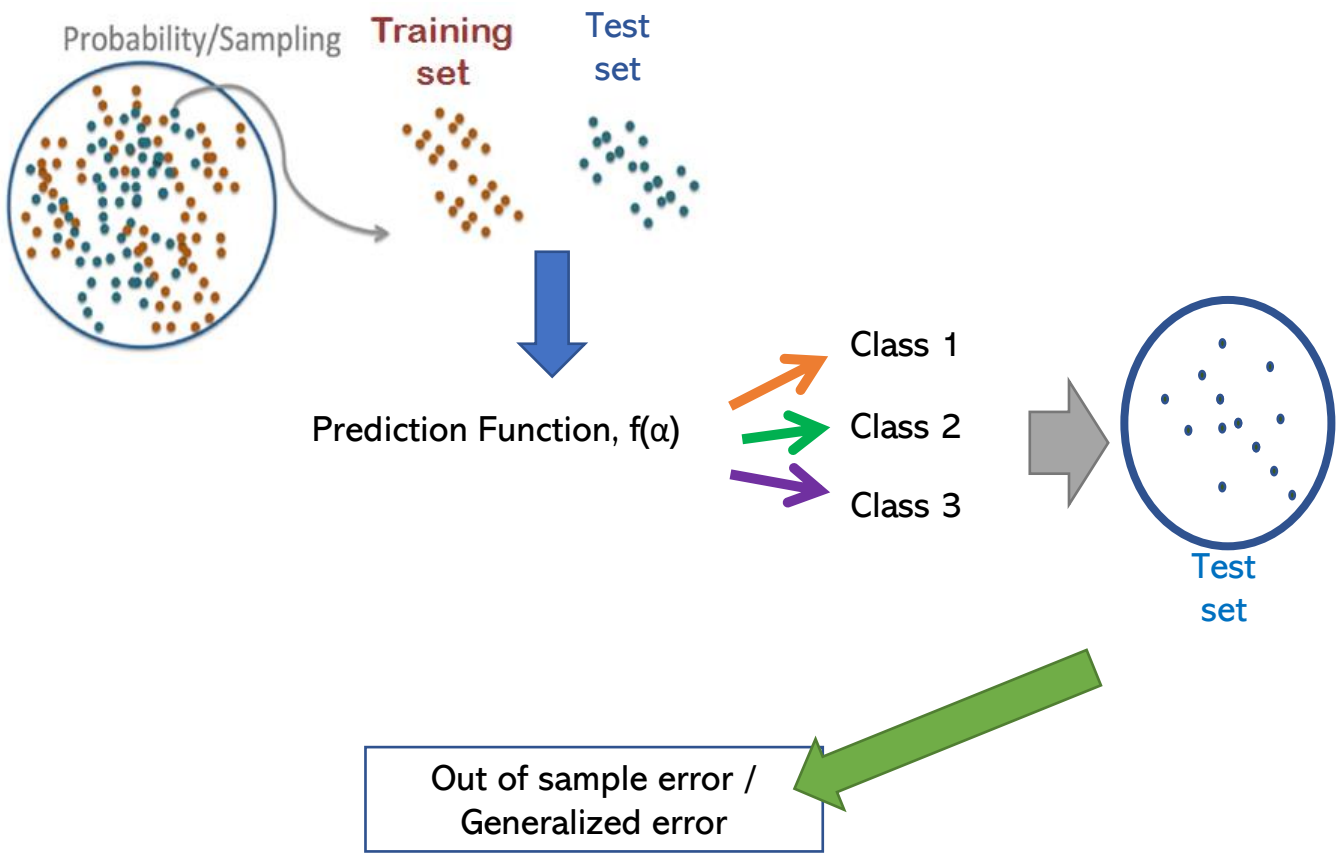
Example: Algorithm

- Supervised Learning: Classification and regression
    - Decision trees Ensembles
    - Support vector machine (SVM)
    - k-NN Linear regression
    - Naive Bayes
    - Neural networks
    - Logistic regression Perceptron
- Unsupervised Learning
    - Hierarchical Clustering
    - k-means Clustering
- Reinforcement Learning
    - Markov Decision Process

Parameters and hyperparameter

- Each of the machine learning algorithms associated with some extra parameters
- Called as hyper parameters
- For example
  - Number of decision trees need to use in the Random Forest Model
  - In classical statistics view : How many components need to use in PLS-DA model ?

Evaluation



Hastie et al., (2009) 2nd Ed., The Elements of Statistical Learning : Data Mining, Inference, and Prediction.